

C Programming & Data Structures

Project - Student Guide

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COURSE DESIGN-PROJECT STUDENT GUIDE

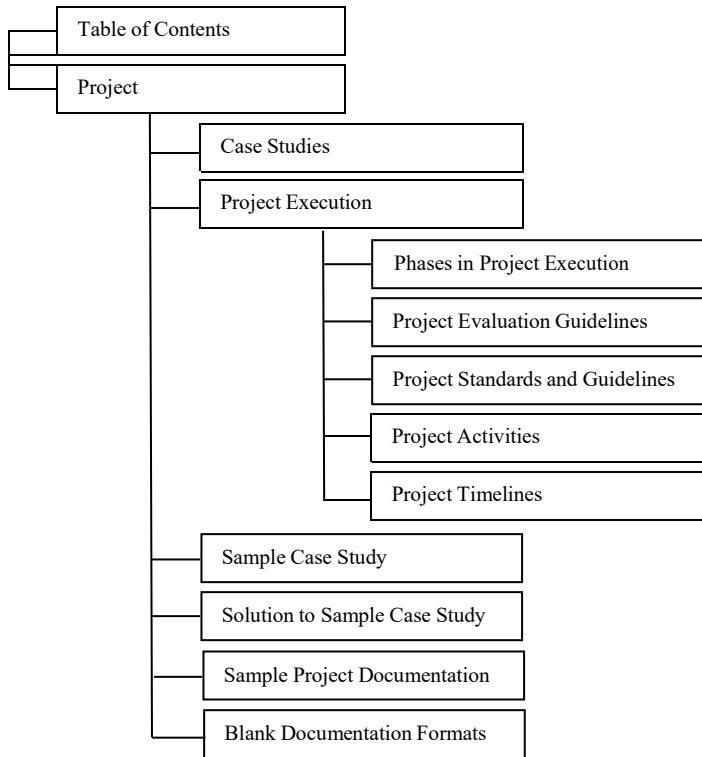
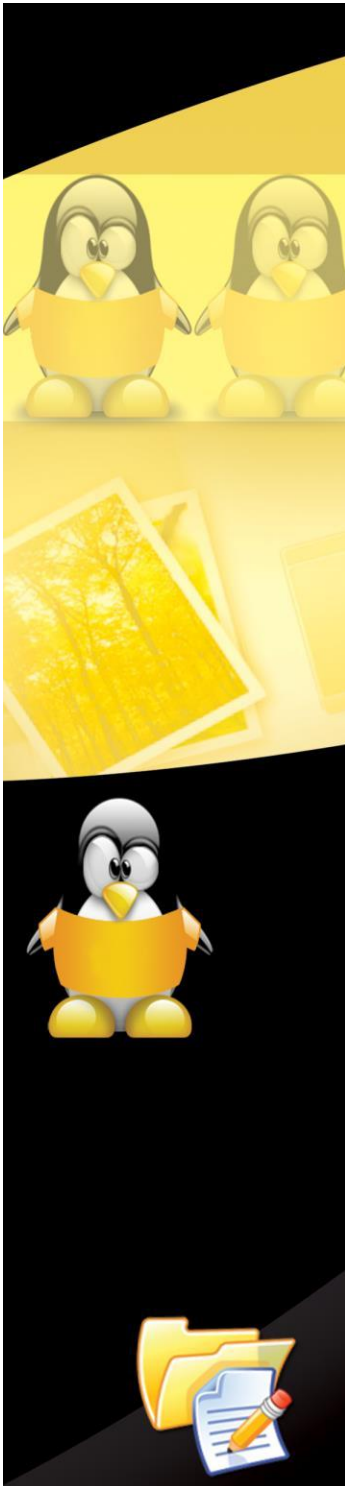


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Project



Case Study 1: JD Health Emergency Triage Priority Queue System

Background 背景

In real hospital emergency rooms, a patient with a minor injury should never be treated before a patient in critical condition — regardless of who arrived first.

Healthcare systems such as JD Health use intelligent triage systems to prioritize patients dynamically based on severity and emergency level.

This project simulates a hospital emergency triage engine using Data Structures.

在真实的医院急诊室里，无论轻伤患者先到，都不应该优先救治危重患者。

京东健康等医疗系统采用智能分诊系统，根据病情严重程度和紧急程度动态地对患者进行优先级排序。

本项目使用数据结构模拟医院急诊分诊引擎。

Envisioned System 构想系统

Students will develop a patient management system using:

学生将使用以下数据结构开发患者管理系统：

- Min Heap Priority Queue
- 最小堆优先队列
- Linked List
- 链表

The system should:

系统应具备：

- Always process the highest priority patient first
- 始终优先处理最高优先级患者
- Dynamically update patient severity
- 动态更新患者病情等级
- Maintain complete patient records
- 维护完整患者记录

Patient Information Design

患者信息设计

Field	Description
-------	-------------

Patient ID	Unique patient number
------------	-----------------------

患者编号	唯一患者编号
------	--------

Name	Patient name
------	--------------

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姓名	患者姓名
Age	Patient age
年龄	患者年龄
Disease	Medical condition
疾病	医疗病情
Severity Level	Emergency priority
严重程度	急诊优先级
Arrival Time	Patient arrival timestamp
到达时间	患者到达时间戳

Required Operations

必须实现的功能

Main Menu

主菜单

- Add Patient
- 添加患者
- View Patients
- 查看患者
- Emergency Queue
- 急诊队列
- Update Severity
- 更新严重程度
- Treat Patient
- 治疗患者
- Search Patient
- 查询患者
- Delete Patient
- 删除患者
- Exit
- 退出系统

Add Patient

添加患者

- Register new patient
- 注册新患者
- Validate unique Patient ID
- 验证患者编号唯一性

- Insert into Min Heap
 - 插入最小堆
 - Add to Linked List registry
 - 添加到链表注册表
-

Emergency Queue

急诊队列

- Display highest priority patient
 - 显示最高优先级患者
 - Heapify after insertion/removal
 - 插入或删除后进行堆调整
-

Update Severity

更新严重程度

- Change patient severity level
 - 修改患者严重程度
 - Rearrange heap automatically
 - 自动重新排列堆结构
-

Treat Patient

治疗患者

- Remove highest priority patient
 - 移除最高优先级患者
 - Update records
 - 更新记录
-

Search Patient

查询患者

Search using:

支持以下查询方式:

- Patient ID
 - 患者编号
 - Patient Name
 - 患者姓名
-

Mandatory Data Structures

必须使用的数据结构

- Min Heap
 - 最小堆
 - Linked List
 - 链表
-

Bonus Features

加分功能

- File Handling
- 文件处理
- GUI Interface
- 图形用户界面
- Doctor Assignment
- 医生分配
- Patient History Tracking
- 患者历史记录跟踪
- Emergency Alerts
- 紧急警报

Case Study 2: SF Express Parcel Routing & Tracking System

Background 背景

Behind every parcel delivered by SF Express is a network of interconnected transit hubs and optimized delivery routes.

每一个由 SF Express 配送的包裹背后，都有一个由中转中心和优化配送路线组成的网络。

Modern logistics systems use graph traversal algorithms to identify the fastest and most efficient delivery paths.

现代物流系统使用图遍历算法来寻找最快和最高效的配送路径。

Envisioned System 构想系统

Students will develop a parcel routing system using:

学生将使用以下数据结构开发包裹路由系统：

- Weighted Graph
- 带权图
- BFS
- 广度优先搜索（BFS）
- DFS
- 深度优先搜索（DFS）
- Stack
- 栈

The system should:

系统应具备：

- Simulate parcel movement between cities
- 模拟包裹在城市之间的运输过程
- Track parcel history
- 跟踪包裹历史记录
- Discover shortest routes

- 查找最短配送路线

Parcel Information Design

包裹信息设计

Field	Description
Parcel ID	Unique parcel number
包裹编号	唯一包裹编号
Sender Name Parcel sender	
发件人姓名 包裹发件人	
Receiver Name Parcel receiver	
收件人姓名 包裹收件人	
Source City Starting location	
起始城市 起始地点	
Destination City Delivery location	
目标城市 配送地点	
Status Current delivery status	
状态 当前配送状态	

Required Operations

必须实现的功能

Main Menu

主菜单

- Add City Hub
- 添加城市节点
- Add Route
- 添加路线

- Register Parcel
 - 注册包裹
 - Find Shortest Route
 - 查找最短路线
 - DFS Traversal
 - DFS 遍历
 - Update Parcel Status
 - 更新包裹状态
 - View Tracking History
 - 查看追踪历史
 - Exit
 - 退出系统
-

Add City Hub

添加城市节点

- Create city nodes
 - 创建城市节点
-

Add Route

添加路线

- Connect cities with weighted routes
 - 使用带权路径连接城市
-

BFS Route Search

BFS 路线搜索

- Find shortest path between cities

- 查找城市之间的最短路径
-

DFS Traversal

DFS 遍历

- Explore all reachable cities
 - 遍历所有可到达城市
-

Parcel Tracking

包裹追踪

Maintain tracking events using Stack:

使用栈维护包裹追踪事件：

- Registered
 - 已注册
 - In Transit
 - 运输中
 - Delivered
 - 已送达
-

Mandatory Data Structures

必须使用的数据结构

- Graph
- 图
- Queue
- 队列
- Stack
- 栈

Bonus Features

加分功能

- GPS Simulation
- GPS 模拟
- Delivery Cost Calculation
- 配送费用计算
- Estimated Delivery Time
- 预计送达时间
- GUI Dashboard
- 图形化仪表板界面

Case Study 3: Alipay Account & Transaction Ledger System

Background 背景

Digital payment systems such as Alipay process millions of transactions rapidly using optimized data structures.

像支付宝这样的数字支付系统会使用优化的数据结构快速处理数百万笔交易。

This project simulates a financial transaction processing engine.
本项目模拟一个金融交易处理引擎。

Envisioned System 构想系统

Students will develop an account management system using:

学生将使用以下数据结构开发账户管理系统：

- Hash Table
- 哈希表
- Doubly Linked List
- 双向链表
- Stack
- 栈

The system should:

系统应具备：

- Provide fast account lookup
- 提供快速账户查询
- Maintain transaction history
- 维护交易历史记录
- Support transaction rollback
- 支持交易回滚

Account Information Design

账户信息设计

Field	Description
Account Number	Unique account ID
账户编号	唯一账户编号

| User Name | Account holder |
| 用户姓名 | 账户持有人 |

Phone Number	Contact number
电话号码	联系电话
Balance	Current balance
余额	当前余额

Required Operations

必须实现的功能

Main Menu

主菜单

- Create Account
 - 创建账户
 - Deposit Money
 - 存款
 - Withdraw Money
 - 取款
 - Transfer Money
 - 转账
 - Search Account
 - 查询账户
 - View Ledger
 - 查看账本
 - Undo Transaction
 - 撤销交易
 - Delete Account
 - 删除账户
 - Exit
 - 退出系统
-

Account Creation

创建账户

- Validate unique account number
 - 验证账户编号唯一性
 - Insert into hash table
 - 插入哈希表
-

Transaction Management

交易管理

Support:

支持以下功能:

- Deposit
- 存款
- Withdrawal
- 取款
- Transfer
- 转账

Transaction Ledger

交易账本

Maintain transaction history using Doubly Linked List.

使用双向链表维护交易历史记录。

Undo Operation

撤销操作

Use Stack to rollback:

使用栈实现回滚操作:

- Last deposit
- 最近一次存款
- Last withdrawal
- 最近一次取款
- Last transfer
- 最近一次转账

Mandatory Data Structures

必须使用的数据结构

- Hash Table
- 哈希表
- Doubly Linked List
- 双向链表
- Stack
- 栈

Bonus Features

加分功能

- Login Authentication

- 登录认证
- PIN Security
- PIN 安全验证
- Monthly Statement Generation
- 月度账单生成
- Fraud Detection Alerts
- 欺诈检测警报

。

Project Execution 项目执行

Phases in Project Execution 项目执行阶段

The project will be carried out in the following phases:

- **Analysis:** Analyze and research the function of the system, and write the analysis solution document.
- **Development:** Develop the system according to the analysis solution document.
- **Testing and debugging:** Test and debug the system to find out the functions that have not been realized or are wrong.
- **Documentation:** finish documents, and submitted on the given date.

项目将分以下几个阶段进行：

- 分析：对系统的功能进行分析和研究，编写业务分析方案。
- 开发：根据业务分析方案进行系统开发。
- 测试和调试：对系统进行测试调试，找出系统尚未实现或者实现有误的功能。
- 文档：编写项目文档，并在指定日期提交。

Project Evaluation Guidelines 项目评估指南

The project is to be evaluated based on the following parameters: 项目将根据以下参数进行评估：

- **Function:** realize all the functions of any system in the guidance document – 30%
功能：实现指导文档中任一系统的所有功能 – 分数占比30%
- **Teamwork:** Demonstrate effective collaboration, clear task distribution, and mutual support during the project development -20%
团队合作：在项目开发过程中展示有效的协作、清晰的任务分配和相互支持20%
- **Document:** The document describe the system clearly and completely, and has the flow chart of related functions-20%
文档：文档中对系统功能进行清晰、完整的描述，并有相关功能流程图-分数占比20%
- **Presentation:** The presentation should be clear and fluent-20%
演示：系统演示运行流畅，语言表达清晰-分数占比20%
- **Additional points :** If you can combine the two systems together, you will get additional points, for example, you can combine Calculator App and Student Management System. In addition, if your project has other highlights, you can also get additional points -10%

加分项：如果您能将两个系统结合起来，即可获得加分，例如，您可以将计算器应用程序和学生管理系统结合起来。此外，如果您的专案还有其他亮点，您还可以获得额外加分（-10%）。

Project Standards and Guidelines 项目标准和指南

This project requires knowledge of all the concepts you have learned. Some of the concepts might not be used

directly but the concepts that you learned will definitely help.

本项目需要运用你所学过的所有知识概念。虽然有些概念可能不会被直接使用，但你所学到的知识一定会对项目开发有所帮助。

Project Activities 项目活动

The students will get four weeks to complete the project. The activities to be performed during this period are: 学生将有四周时间完成该项目。在此期间需要完成的活动包括：

1. Analyze the given case study to identify the requirements. 分析给定的案例来确定需求
2. Develop strategies to meet the identified requirements. In addition, mention the rationale for suggesting a specific strategy. 制定策略满足制定的需求，并给出具体的理由。
3. Develop the required application. 开发所需的应用程序。
4. Test and debug the application. 测试和调试应用程序。
5. Document the project. 编写项目文档。
6. Submit the documentation to the faculty. 提交文档给老师。
7. Present the project to the faculty. 向老师演示项目。

The duration of the project is four weeks. These two weeks can be divided phase wise according to the following break-up. 项目周期为四周。这四周可以按照以下阶段进行划分：

<i>Activity</i>	<i>Duration</i>
<i>Analysis</i> 分析	<i>Six days</i> 六天
<i>Development</i> 开发	<i>Two weeks</i> 两周
<i>Testing and debugging</i> 测试	<i>Four days</i> 四天
<i>Documentation</i> 文档	<i>Four days</i> 四天

The Duration of the Activity
活动持续时间